A dynamic program chooses grade boundaries

Fixed order A, B, C, D.
w = (3/5, 2/5), τ = 1/4; M = 6.

A. Signed savings $W(a, b)$

Left cut boundary a \ Right cut boundary b	1	2	3	4
0	**0**	0	0	−9/20
1	—	0	**+3/20**	−3/20
2	—	—	0	−3/20
3	—	—	—	**0**

W covers positions a+1 through b;
signed savings may be negative.

B. Prefix savings $V_h(c)$

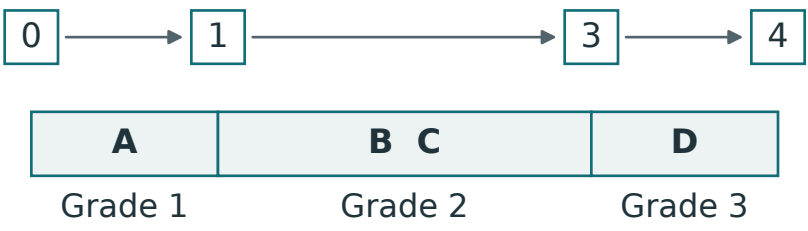
Nonempty blocks h \ Prefix length c	0	1	2	3	4
0	**0**	−∞	−∞	−∞	−∞
1	−∞	**0**	0	0	−9/20
2	−∞	−∞	0	**+3/20**	0
3	−∞	−∞	−∞	0	**+3/20**
4	−∞	−∞	−∞	−∞	0

−∞ is infeasible; $V_0(0)=0$ is auxiliary.
Outlines select the three-block solution.

$$V_h(c) = \max_b \{V_{h-1}(b) + W(b, c)\}$$

$h - 1 \le b < c$; unnormalized maximum saving.

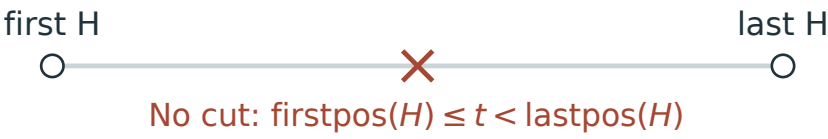
C. Recover the three-block report



g = (1,2,2,3); N = 5. Saving = 3/20.
Risk reduction = 1/40; R = 7/40 − (3/20)/6 = 3/20.

Exactly 4: risk 7/40. At most 4: risk 3/20.

D. Guarded neutral extension



Full active-profile equality class H
stays in one grade. Neutral loss/baseline;
original units and pair multiplicities.
Symbolic: not this tie-free example;
no package DP support is asserted.